A Workflow for Protein Function Discovery

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ABSTRACT

A workflow is presented for identifying from within a large set of proteins, those that have been identified in the literature as being associated with one of a set of concepts related to cancer pain. The workflow uses the programming utilities available through NCBI for accessing their linked databases from within a Matlab programming environment to replace a formerly time-intensive manual process with a much more efficient automated process.

INTRODUCTION

In a study undertaken to define the repertoire of peptides and proteases that produce pain in the cancer microenvironment of oral cancer patients, a protein chemist identifies proteins in samples taken from oral cancer patients by a surgeon-scientist. The protein chemist uses a discovery approach that produces several hundreds of candidate molecules that require screening and validation (Hardt et al. 2011). Likely candidates for involvement in the pain process are selected based on published studies of these proteins. Once chosen, proteins are then either isolated from the sample or synthesized, so that their nociceptive (i.e. pain-producing) capacity can be determined in a preclinical model, so it is critical to choose the best candidates for this further testing. The search for literature on protein function had been done manually, and this cumbersome and time-intensive approach presented an impediment in the study. A team of librarians was brought in to collaborate in the process in order to help overcome this information management challenge.

METHODS

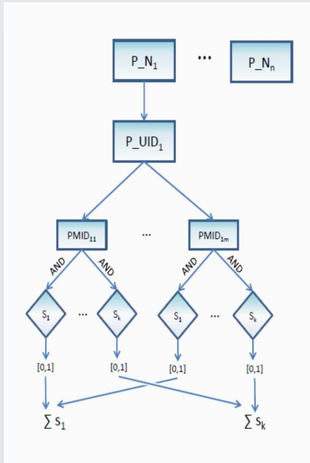
The method of literature searching being used by the researchers was keyword searching in PubMed and Google Scholar, which was too time consuming to allow for a thorough search, and did not make use of searching features that could improve the effectiveness of the search. An automated search methodology was introduced that programmatically accessed the linked NCBI databases through the use of the Entrez programming utilities (NCBI 2010). Databases could then be automatically queried and linkages between the databases could be exploited.

RESULTS

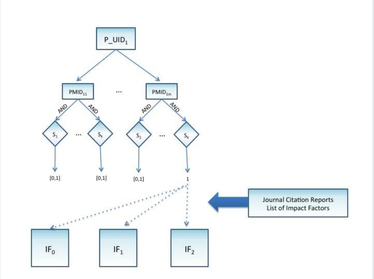
An automated workflow was established to discover which of the identified proteins were associated with one of a number of terms related to cancer pain, or to pain more generally, as well as to identify those proteins for which there was no published literature regarding their function. The workflow was set up within the Matlab programming environment, through calls to the NCBI programming utilities, as follows:

- Protein names used to retrieve protein unique identifiers (UIDs).
- Protein UIDs used to retrieve linked papers from PubMed
- Papers searched for each cancer and/or pain related term

The final output was an Excel table of size N x (2*M), where N was the number of proteins and M was the number of concepts for which the search was done. For each protein, the total number of papers containing the concept was output, as well as a column containing the PubMed IDs of those papers.



Workflow from protein name (P_N) through protein identifier (P_UID) to relevant papers (PMID) that were then searched for each relevant search term (S) to establish a measure of known associations of the protein with the concept (∑S).



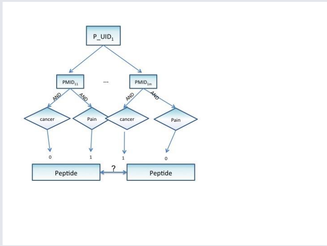
Papers (PMIDs) that are identified as containing one of the search terms (S) would be binned according to the value of the impact factor (IF) of that journal.

NEXT STEPS

Two further refinements are currently being implemented.

- Filtering results by impact factor
- Finding proteins with concepts joined through common peptide/protease

The first of these is to use impact factor as a proxy for reproducibility of results, to aid in decisions by the researchers as to which proteins to use for preclinical testing. The second is an expansion of the search to locate proteins that are not implicated in cancer pain, but that are implicated in both separate concepts where a common peptide or protease is also referenced.



Papers (PMIDs) linked to a given protein (P_UID) that return a match for search terms related to one concept (e.g. cancer) will be checked to see if they are associated with a peptide or protease that is also associated with papers related to a different concept (e.g. pain).

CONCLUSIONS

This technique provides a practical means of doing a more comprehensive survey of literature on known functions of a large list of identified proteins. Further enhancements of the method will provide additional information to allow researchers to choose the best candidates for use in preclinical models.

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